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Inventor(s)

TAKASHIMA; Naoyuki et al.

FIRST JOINT AND PIPE JOINT

Abstract

A first joint has a central axis and extends in an axial direction along the central axis. The first joint includes a first joint body, an axially facing portion, an actuation portion, and a mounting portion. The first joint body is coupled to the second joint along the central axis. The axially facing portion axially opposes a protruding portion protruding from an outer peripheral surface of the second joint, from a front end side in an axial direction of the first joint body. The actuation portion is connected to the axially facing portion and is actuatable along an axis intersecting direction intersecting the central axis. The mounting portion includes an actuation portion restricting portion restricting movement of the actuation portion in a state where the axially facing portion and the protruding portion oppose each other.

Inventors: TAKASHIMA; Naoyuki (Kyoto, JP), TAMAOKA; Takehito (Kyoto, JP), TOKESHI; Toshihiko (Kyoto, JP)

Applicant: NIDEC CORPORATION (Kyoto, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This is a Continuation of PCT Application No. PCT/JP2023/042758, filed on Nov. 29, 2023, with priority being claimed from Japanese Applications No. 2022-192116, filed Nov. 30, 2022, and 2023-200413, filed Nov. 28, 2023, the entire disclosures of which are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to first joints and pipe joints.

2. Background

[0003] In a conventional joint device, a user manually operates the actuation portion to quickly connect and disconnect the joint body with a portion of the fluid transport device. The actuation portion reciprocates with respect to the outer surface of the joint body by the action of the biasing member (for example, US 2005/0001425 A).

[0004] Specifically, an insert assembly is inserted into the joint body. In this case, the tapered surface of the insert assembly locks the insert assembly with respect to the joint body. Further, the lock is released by pushing down the actuation portion, and the insert assembly can be detached and removed from the joint body.

SUMMARY

[0005] However, in the conventional joint device, for example, when the actuation portion comes into contact with another member, the lock of the insert assembly may be released due to malfunction of the actuation portion.

[0006] Example embodiments of the present disclosure have been made in view of the above problem, and provide first joints and pipe joints each capable of reducing or preventing malfunction of an actuation portion.

[0007] A first joint according to an example embodiment of the present disclosure includes a central axis and extends axially along the central axis. The first joint includes a first joint body, an axially facing portion, an actuation portion, and a mounting portion. The first joint body is coupled to the second joint along the central axis. In a state where the first joint body is coupled to the second joint, the axially facing portion axially opposes a protruding portion protruding from an outer peripheral surface of the second joint, from a front end side in an axial direction of the first joint body. The actuation portion is connected to the axially facing portion and is actuatable along an axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction. The mounting portion is mounted on the outer peripheral surface of the first joint body. The mounting portion includes an actuation portion restricting portion to restrict movement of the actuation portion in a state where the axially facing portion and the protruding portion face each other.

[0008] An example embodiment of a first joint of the present disclosure has a central axis and extends axially along the central axis. The first joint includes a first joint body, an axially facing portion, an actuation portion, and a mounting portion. The first joint body is coupled to the second joint along the central axis. In a state where the first joint body is coupled to the second joint, the axially facing portion axially opposes a protruding portion protruding from the outer peripheral

surface of the second joint, from the front end side in the axial direction of the first joint body. The actuation portion is connected to the axially facing portion and is actuatable along the axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction. The mounting portion is mounted on the outer peripheral surface of the first joint body. The mounting portion includes an intersecting direction facing portion. In a state where the axially facing portion and the protruding portion oppose each other, the intersecting direction facing portion opposes the actuation portion in the axis intersecting direction between the first joint body and the actuation portion.

[0009] An example embodiment of a pipe joint of the present disclosure includes the first joint and the second joint described above.

[0010] According to example embodiments of the present disclosure, malfunction of the actuation portion can be reduced or prevented.

[0011] The above and other elements, features, steps, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a perspective view illustrating a second joint according to a first example embodiment of the present disclosure.

[0013] FIG. 2 is a cross-sectional view taken along line II-II of FIG. 1.

[0014] FIG. 3 is a perspective view illustrating a first joint according to the first example embodiment of the present disclosure

[0015] FIG. 4 is an exploded perspective view illustrating the first joint according to the first example embodiment of the present disclosure.

[0016] FIG. 5 is a cross-sectional view taken along line V-V of FIG. 3.

[0017] FIG. 6 is a cross-sectional view illustrating a state in which an actuation portion of the first joint according to the first example embodiment of the present disclosure is located at an open position.

[0018] FIG. 7 is a cross-sectional view illustrating a state in which the first joint and the second joint according to the first example embodiment of the present disclosure are connected.

[0019] FIG. 8 is a plan view illustrating the first joint according to the first example embodiment of the present disclosure.

[0020] FIG. 9 is a cross-sectional view taken along line IX-IX of FIG. 8.

[0021] FIG. 10 is a plan view illustrating another state of the first joint according to the first example embodiment of the present disclosure.

[0022] FIG. 11 is a cross-sectional view taken along line XI-XI of FIG. 10.

[0023] FIG. 12 is a plan view illustrating a mounting portion according to the first example embodiment of the present disclosure.

[0024] FIG. 13 is a perspective view illustrating the mounting portion according to the first example embodiment of the present disclosure.

[0025] FIG. 14 is a rear view illustrating the mounting portion according to the first example embodiment of the present disclosure.

[0026] FIG. 15 is a front view illustrating the first joint according to the first example embodiment of the present disclosure.

[0027] FIG. 16 is a perspective view illustrating a first joint according to a modification of the first example embodiment of the present disclosure.

[0028] FIG. **17A** is a diagram illustrating a first position of a mounting portion according to the present modification.

[0029] FIG. **17B** is a diagram illustrating a second position of the mounting portion according to the present modification.

[0030] FIG. **18** is a plan view illustrating the first joint when the mounting portion according to the present modification is located at the first position.

[0031] FIG. **19** is a plan view illustrating the first joint when the mounting portion according to the present modification is located at the second position.

[0032] FIG. **20** is a cross-sectional view illustrating a first joint according to a second example embodiment of the present disclosure.

[0033] FIG. **21** is a cross-sectional view illustrating the first joint according to the second example embodiment of the present disclosure.

[0034] FIG. **22** is a cross-sectional view illustrating the first joint according to the second example embodiment of the present disclosure.

[0035] FIG. **23** is a cross-sectional view illustrating a first joint according to a third example embodiment of the present disclosure.

[0036] FIG. **24** is a cross-sectional view illustrating the first joint according to the third example embodiment of the present disclosure.

[0037] FIG. **25** is a cross-sectional view illustrating a first joint according to a fourth example embodiment of the present disclosure.

[0038] FIG. **26** is a cross-sectional view illustrating the first joint according to the fourth example embodiment of the present disclosure.

[0039] FIG. **27** is a cross-sectional view illustrating the first joint according to the fourth example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0040] Hereinafter, example embodiments of the present disclosure will be described with reference to the drawings. Note that in the drawings, the same or corresponding parts will be denoted by the same reference symbols and description of such parts will not be repeated. In the drawings, the X, Y, and Z axes of the three-dimensional orthogonal coordinate system are described as appropriate for ease of understanding.

[0041] In the present description, a direction parallel to a central axis **AX1** (for example, FIG. **3**) of a first joint is referred to as an “axial direction **AD1**”. Further, a direction orthogonal to the central axis **AX1** is referred to as a “radial direction **RD1**”. The “radial direction **RD1**” may be any direction as long as the direction is a direction orthogonal to the central axis **AX1**, and is not particularly limited. Further, a direction along an arc around the central axis **AX1** is referred to as a “circumferential direction **CD1**”.

[0042] Further, a direction parallel to a central axis **AX2** (for example, FIG. **1**) of a second joint is referred to as an “axial direction **AD2**”. Further, a direction orthogonal to the central axis **AX2** is referred to as a “radial direction **RD2**”. The “radial direction **RD2**” may be any direction as long as the direction is a direction orthogonal to the central axis **AX2**, and is not particularly limited. Further, a direction along an arc around the central axis **AX2** is referred to as a “circumferential direction **CD2**”.

[0043] Further, in the present description, a “parallel direction” includes a substantially parallel direction, and an “orthogonal direction” includes a substantially orthogonal direction. Furthermore, in the present description, an “annular shape”, a “cylindrical shape”, an “annular shape”, a “tubular shape”, a “circular shape”, and a “wavy shape” do not represent a shape in a strict sense, and include, for example, a shape that can realize a function of the first joint or the second joint in the present disclosure.

First Example Embodiment

[0044] A first joint **100** and a second joint **200** according to a first example embodiment of the

present disclosure will be described with reference to FIGS. 1 to 15. First, the second joint **200** will be described with reference to FIGS. 1 and 2.

[0045] FIG. 1 is a perspective view illustrating the second joint **200** according to the example embodiment of the present disclosure. As illustrated in FIG. 1, the second joint **200** has the central axis AX2. The second joint **200** extends in the axial direction AD2 along the central axis AX2. The second joint **200** is a male joint. In the present example embodiment, the “front” of the second joint **200** indicates a side to be inserted into the first joint **100** (FIG. 3). “Rear” of the second joint **200** indicates a side to be inserted into a tube TB2.

[0046] The second joint **200** has a protruding portion **204**. Specifically, the second joint **200** includes a plug part **201** and an insertion part **202**. The plug part **201** and the insertion part **202** are coupled in the axial direction AD2. The plug part **201** includes a plug body **203**, the protruding portion **204**, and a coupling body **205**.

[0047] The plug body **203** has a cylindrical shape. The protruding portion **204** protrudes from an outer peripheral surface of the second joint **200**. Specifically, the protruding portion **204** protrudes toward the outer side in the radial direction RD2 from the outer peripheral surface of the plug body **203**. The protruding portion **204** extends along the circumferential direction CD2. The protruding portion **204** has an annular shape. The protruding portion **204** extends with respect to the central axis AX2 from a front end **203a** toward a rear end **206a** of the second joint **200** in the axial direction AD2. That is, the diameter of the protruding portion **204** increases from the front end **203a** toward the rear end **206a** of the second joint **200**. The protruding portion **204** is located on a rear end portion **203b** side in the axial direction AD2 of the plug body **203**.

[0048] The coupling body **205** is connected to the rear end portion **203b** of the plug body **203**. The coupling body **205** and the plug body **203** are constituted as a single member (integrally molded product). The coupling body **205** has a cylindrical shape.

[0049] The insertion part **202** is inserted into the tube TB2. Specifically, the insertion part **202** includes an insertion body **206**, a coupling body **207**, and a plurality of protrusions **212**. The insertion body **206** is inserted into the tube TB2. The insertion body **206** has a cylindrical shape. The protrusion **212** protrudes toward the outer side in the radial direction RD2 from an outer peripheral surface of the insertion body **206**. The protrusion **212** has an annular shape. The diameter of the protrusion **212** increases from the rear end **206a** toward the front end **203a** in the axial direction AD2 of the second joint **200**.

[0050] The coupling body **207** is connected to the rear end portion **206b** in the axial direction AD2 of the insertion body **206**. The coupling body **207** and the insertion body **206** are constituted as a single member (integrally molded product). The coupling body **207** has a cylindrical shape.

[0051] The coupling body **205** of the plug part **201** is fitted into the coupling body **207** of the insertion part **202**. As a result, the plug part **201** and the insertion part **202** are coupled.

[0052] FIG. 2 is a cross-sectional view taken along line II-II of FIG. 1. FIG. 2 illustrates a state in which the second joint **200** is separated from the first joint **100**. That is, a state in which the second joint **200** is not coupled to the first joint **100** is illustrated. As illustrated in FIG. 2, the plug part **201** further includes a plug valve **208**, an elastic portion **209**, a seal member **210**, a seal member **211**, and a flow path **213**. The insertion part **202** further includes a flow path **214**. The flow paths **213** and **214** extend along the axial direction AD2. The front end in the axial direction AD2 of the flow path **214** is connected to the rear end in the axial direction AD2 of the flow path **213**. That is, the flow path **213** and the flow path **214** are connected.

[0053] The plug valve **208** is arranged in the flow path **213** on the front end **203a** side of the plug body **203**. The plug valve **208** is pushed from the rear end portion **203b** side to the front end **203a** side of the plug body **203** by the elastic portion **209**. The plug valve **208** closes the flow path **213** on the front end **203a** side of the plug body **203**. The seal member **210** is attached to the outer peripheral surface of the plug valve **208**. The seal member **210** is an elastic body and has an annular shape. Specifically, the seal member **210** seals the flow path **213** on the front end **203a** side

of the plug body **203**.

[0054] The elastic portion **209** is arranged in the flow path **213**. The elastic portion **209** has elasticity. The elastic portion **209** is, for example, a compression coil spring. The seal member **211** is attached to the outer peripheral surface of the coupling body **205**. The seal member **211** is an elastic body and has an annular shape. The seal member **211** seals between the outer peripheral surface of the coupling body **205** and the inner peripheral surface of the coupling body **207**.

[0055] Next, the first joint **100** will be described with reference to FIGS. **3** to **6**. FIG. **3** is a perspective view illustrating the first joint **100** according to the present example embodiment. FIG. **4** is an exploded perspective view illustrating the first joint **100**.

[0056] As illustrated in FIGS. **3** and **4**, the first joint **100** has the central axis AX1. The first joint **100** extends in the axial direction AD1 along the central axis AX1. The first joint **100** is a female joint. The first joint **100** is coupled to the second joint **200** (FIG. **1**) along the central axis AX1. In the present example embodiment, the “front” of the first joint **100** indicates a side on which the second joint **200** is inserted. The “rear” of the first joint **100** indicates a side to be inserted into a tube TB1.

[0057] The first joint **100** includes a first joint body **1**, a mounting portion **3**, an actuation portion **71**, and an elastic portion **11**. Specifically, the first joint **100** includes an actuation portion **7**. The actuation portion **7** has the actuation portion **71**. The first joint body **1** extends along the axial direction AD1. The first joint body **1** is coupled to the second joint **200** (FIG. **1**) along the central axis AX1. The mounting portion **3** is mounted on the outer peripheral surface of the first joint body **1**. Details of the mounting portion **3** will be described later. The actuation portion **7** is operated by an operator to switch the coupling state between the first joint body **1** and the second joint **200** between a locked state and an unlocked state. The locked state indicates a state in which coupling between the first joint body **1** and the second joint **200** cannot be released. The unlocked state indicates a state in which coupling between the first joint body **1** and the second joint **200** can be released. The elastic portion **11** supports the actuation portion **7**. As an example, the actuation portion **7** (actuation portion **71**) and the elastic portion **11** constitute a push button.

[0058] Specifically, the first joint body **1** includes a socket part **5** and an insertion part **9**. The socket part **5** and the insertion part **9** are connected in the axial direction AD1. The plug part **201** (FIG. **1**) of the second joint **200** is inserted into the socket part **5**. The mounting portion **3** is mounted on the outer peripheral surface of the socket part **5**.

[0059] The socket part **5** has a socket body **50**. The plug body **203** (FIG. **1**) of the second joint **200** is inserted into the socket body **50**. The socket body **50** has a cylindrical shape.

[0060] The socket body **50** has an opening **51**. The opening **51** has a circular shape. The opening **51** is located on a front end **5a** side of the socket body **50**. The opening **51** is open in the axial direction AD1.

[0061] With continued reference to FIGS. **3** and **4**, the configuration related to the actuation portion **7** and the actuation portion **7** will be described. The socket body **50** further includes a through hole **52** and a recess **53**. The through hole **52** penetrates the socket body **50** in an axis intersecting direction Da. The axis intersecting direction Da intersects the central axis AX1. In the present example embodiment, the axis intersecting direction Da is orthogonal to the central axis AX1. The actuation portion **7** is disposed in the through hole **52** along the axis intersecting direction Da. The recess **53** is recessed in the axis intersecting direction Da with respect to the outer peripheral surface of socket body **50**. The elastic portion **11** is disposed in the recess **53** along the axis intersecting direction Da. The elastic portion **11** has elasticity. The elastic portion **11** is, for example, a compression coil spring. The elastic portion **11** extends and contracts in the axis intersecting direction Da. The elastic portion **11** is disposed between the actuation portion **71** of the actuation portion **7** and the bottom surface **53a** of the recess **53**. The elastic portion **11** supports the actuation portion **71**.

[0062] The socket body **50** further includes a pair of contact portions **54**. In FIGS. **3** and **4**, only

one contact portion **54** appears. The actuation portion **7** further includes a pair of connecting portions **72**, a pair of stoppers **73**, an opening **74**, and an axially facing portion **75**.

[0063] One side **D1** and the other side **D2** of the axis intersecting direction **Da** will be defined. One side **D1** indicates a direction (first direction) from the axially facing portion **75** toward the actuation portion **71** in a state where the actuation portion **7** is disposed in the through hole **52**. The other side **D2** indicates a direction (second direction) from the actuation portion **71** toward the axially facing portion **75** in a state where the actuation portion **7** is disposed in the through hole **52**.

[0064] The pair of connecting portions **72** extends along the axis intersecting direction **Da**. The pair of connecting portions **72** connects the actuation portion **71** and the axially facing portion **75**. That is, the actuation portion **71** is connected to the axially facing portion **75** by the pair of connecting portions **72**. The connecting portion **72** has a columnar shape. The actuation portion **71**, the pair of connecting portions **72**, and the axially facing portion **75** define the opening **74**. The opening **74** opens in the axial direction **AD1**.

[0065] The pair of stoppers **73** is disposed at both ends of the axially facing portion **75** in a direction **Db**. The direction **Db** indicates a direction orthogonal to the axis intersecting direction **Da** and the axial direction **AD1**. As illustrated in FIG. 3, in a state where the actuation portion **7** is disposed in the through hole **52**, the pair of stoppers **73** faces the pair of contact portions **54** in the axis intersecting direction **Da**. Specifically, in a state where the stopper **73** and the contact portion **54** face each other, the contact portion **54** is located on the one side **D1** of the stopper **73**, and the stopper **73** is located on the other side **D2** of the contact portion **54**. The pair of contact portions **54** face each other in the direction **Db**. The through hole **52** is provided between the pair of contact portions **54**.

[0066] In a state where the actuation portion **7** is disposed in the through hole **52**, the elastic portion **11** pushes the actuation portion **71** toward the one side **D1** in the axis intersecting direction **Da**. In this case, the stopper **73** of the actuation portion **7** comes into contact with the contact portion **54** in the axis intersecting direction **Da**. Therefore, the stopper **73** prevents the actuation portion **7** from coming out of the through hole **52** toward the one side **D1** in the axis intersecting direction **Da**. That is, actuation portion **7** is held by the socket body **50** while being disposed in through hole **52**. The position of the actuation portion **7** when the stopper **73** is in contact with the contact portion **54** may be referred to as an “initial position”.

[0067] On the other hand, when a force is applied to the actuation portion **7** toward the other side **D2** in the axis intersecting direction **Da** in a state where the actuation portion **7** is disposed in the through hole **52**, the actuation portion **7** moves from the initial position toward the other side **D2** in the axis intersecting direction **Da** against the elastic force of the elastic portion **11**. When the force on the actuation portion **7** is released, due to the elastic force of the elastic portion **11**, the actuation portion **7** moves toward the one side **D1** in the axis intersecting direction **Da** until the stopper **73** comes into contact with the contact portion **54**. That is, the actuation portion **7** returns to the initial position.

[0068] Subsequently, referring to FIGS. 3 and 4, the insertion part **9** is inserted into the tube **TB1**. Specifically, the insertion part **9** includes an insertion body **91**, a coupling body **92**, and a plurality of protrusions **93**. The insertion body **91** is inserted into the tube **TB1**. The insertion body **91** has a cylindrical shape. The protrusion **93** protrudes outward in the radial direction **RD1** from an outer peripheral surface of the insertion body **91**. The protrusion **93** has an annular shape. The diameter of the protrusion **93** increases from the rear end **91a** toward the front end **5a** in the axial direction **AD1** of the first joint **100**.

[0069] The coupling body **92** is coupled to the socket part **5**. The coupling body **92** is connected to the rear end portion **91b** in the axial direction **AD1** of the insertion body **91**. The coupling body **92** and the insertion body **91** are constituted as a single member (integrally molded product). The coupling body **92** has a cylindrical shape.

[0070] Next, the internal structure of the first joint **100** will be described with reference to FIG. 5.

FIG. 5 is a cross-sectional view taken along line V-V of FIG. 3. FIG. 5 illustrates a state in which the first joint **100** is separated from the second joint **200**. That is, a state in which the first joint **100** is not coupled to the second joint **200** is illustrated.

[0071] As illustrated in FIG. 5, the socket part **5** further includes a coupling body **55**, a socket valve **56**, a stem portion **57**, an elastic portion **58**, seal members **59**, **60**, **61**, and **65**, a flow path **62**, and a plug arrangement portion **63**. In the plug arrangement portion **63**, the plug body **203** (FIG. 1) is arranged. Specifically, the plug arrangement portion **63** has a plug arrangement space **67**. Then, the plug body **203** is arranged in the plug arrangement space **67**. Further, the plug arrangement portion **63** has a protrusion arrangement portion **66**. The protruding portion **204** (FIG. 1) of the plug body **203** is arranged in the protrusion arrangement portion **66**. The insertion part **9** further includes a flow path **94**. The first joint body **1** further includes an overhanging portion **64**. Specifically, the socket body **50** of socket part **5** has the overhanging portion **64**. Details of the overhanging portion **64** will be described later.

[0072] The flow paths **94** and **62** and the plug arrangement space **67** extend along the axial direction **AD1**. A front end in the axial direction **AD1** of the flow path **94** is connected to the rear end in the axial direction **AD1** of the flow path **62**. That is, the flow path **94** and the flow path **62** are connected to each other. Further, the front end in the axial direction **AD1** of the flow path **62** is connected to the rear end in the axial direction **AD1** of the plug arrangement space **67**. That is, the flow path **62** and the plug arrangement space **67** are connected to each other. The front end in the axial direction **AD1** of the plug arrangement space **67** is connected to the opening **74**. That is, the plug arrangement space **67** is connected to the opening **74**. Further, the opening **74** is connected to the opening **51** in the axial direction **AD1**.

[0073] The coupling body **55** is connected to the overhanging portion **64** of the socket body **50**. The overhanging portion **64** is positioned at a rear end portion in the axial direction **AD1** of the socket body **50**. The coupling body **55** and the socket body **50** are constituted as a single member (integrally molded product). The coupling body **55** has a cylindrical shape.

[0074] The coupling body **55** of the socket part **5** is fitted to the coupling body **92** of the insertion part **9**. As a result, the socket part **5** and the insertion part **9** are coupled.

[0075] The socket valve **56** has a cylindrical shape. The socket valve **56** is arranged in a rear end portion of the plug arrangement space **67**. The socket valve **56** is pushed by the elastic portion **58** from the overhanging portion **64** side to the front end **5a** side of the socket body **50**. The socket valve **56** closes the rear end portion of the plug arrangement space **67**. The seal member **65** is disposed in the plug arrangement space **67**. The seal member **65** is an elastic body and has an annular shape. When the seal member **65** comes into contact with the outer peripheral surface of the socket valve **56**, the rear end portion of the plug arrangement space **67** is sealed.

[0076] The stem portion **57** extends along the axial direction **AD1**. A front end portion **57a** in the axial direction **AD1** of the stem portion **57** is arranged in the rear end portion of the plug arrangement space **67**. In the example of FIG. 5, the front end portion **57a** is arranged inside the socket valve **56**. The seal member **60** is attached to the outer peripheral surface of the front end portion **57a**. The seal member **60** is an elastic body and has an annular shape. Specifically, the seal member **60** seals a space between the outer peripheral surface of the stem portion **57** and the inner peripheral surface of the socket valve **56**. Further, the seal member **61** is arranged in the plug arrangement space **67**. The seal member **61** is disposed closer to the front end **5a** of the socket body **50** than the seal member **65**.

[0077] The elastic portion **58** is arranged in the flow path **62**. The elastic portion **58** has elasticity. The elastic portion **58** is, for example, a compression coil spring. The seal member **59** is attached to the outer peripheral surface of the coupling body **55**. The seal member **59** is an elastic body and has an annular shape. The seal member **59** seals a space between the outer peripheral surface of the coupling body **55** and the inner peripheral surface of the coupling body **92**.

[0078] Next, the actuation portion **7** will be described with reference to FIGS. 5 and 6. FIG. 5

shows the actuation portion 7 when the axially facing portion 75 is located at the initial position Z0. As illustrated in FIG. 5, in a state where the actuation portion 7 (actuation portion 71) is pressed by the elastic portion 11 to the one side D1 of the axis intersecting direction Da and is stationary, the axially facing portion 75 is located at the initial position Z0. At the initial position Z0, the axially facing portion 75 faces the protrusion arrangement portion 66 in the axial direction AD1. That is, at the initial position Z0, the axially facing portion 75 faces the protrusion arrangement portion 66 in the axial direction AD1.

[0079] FIG. 6 illustrates the actuation portion 7 when the axially facing portion 75 is located at the open position Z1. As illustrated in FIG. 6, when an external force is applied to the actuation portion 7 (actuation portion 71) toward the other side D2 in the axis intersecting direction Da, the actuation portion 7 (actuation portion 71) is pressed against the elastic force of the elastic portion 11 toward the other side D2 in the axis intersecting direction Da and stands still. As a result, the axially facing portion 75 moves from the initial position Z0 to the open position Z1 and stands still. At the open position Z1, the axially facing portion 75 is located on the other side D2 in the axis intersecting direction Da with respect to the protrusion arrangement portion 66. Therefore, at the open position Z1, the axially facing portion 75 does not face the protrusion arrangement portion 66 in the axial direction AD1. As a result, the front side in the axial direction AD1 of the protrusion arrangement portion 66 is opened at the open position Z1. The “external force” is applied to the actuation portion 7 (actuation portion 71) by an operator, for example. For example, the operator presses the actuation portion 71 with a finger.

[0080] Next, a state in which the first joint 100 and the second joint 200 are coupled will be described with reference to FIG. 7. FIG. 7 is a cross-sectional view illustrating a state in which the first joint 100 and the second joint 200 are coupled. In a state where the first joint 100 and the second joint 200 are coupled to each other, the central axis AX1 of the first joint 100 and the central axis AX2 of the second joint 200 coincide with each other. Therefore, the axial direction AD1 and the axial direction AD2 coincide with each other, the radial direction RD1 and the radial direction RD2 coincide with each other, and the circumferential direction CD1 and the circumferential direction CD2 coincide with each other.

[0081] As illustrated in FIG. 7, the first joint 100 and the second joint 200 constitute a pipe joint 300. That is, the pipe joint 300 includes the first joint 100 and the second joint 200.

[0082] When the plug body 203 is inserted into the socket body 50, the front end 203a in the axial direction AD1 of the plug body 203 and a front end 56a in the axial direction AD1 of the socket valve 56 come into contact with each other. Therefore, the socket valve 56 moves from one side D11 toward the other side D12 in the axial direction AD1 by the plug body 203. At the same time, when the plug body 203 is inserted into the socket body 50, the front end portion 57a in the axial direction AD1 of the stem portion 57 and a front end portion 280b in the axial direction AD1 of the plug valve 208 come into contact with each other. Therefore, the plug valve 208 moves from the other side D12 to the one side D11 in the axial direction AD1. As a result, the flow path 213 of the plug part 201 and the flow path 62 of the socket part 5 are connected. Therefore, liquid flows between the first joint 100 and the second joint 200.

[0083] Subsequently, with reference to FIG. 7, a case where the coupling state between the first joint 100 and the second joint 200 is switched between the locked state and the unlocked state will be described. In a state where the first joint 100 and the second joint 200 are coupled, the plug body 203 is disposed in the plug arrangement portion 63 (plug arrangement space 67) of the socket body 50. In this case, the protruding portion 204 of the plug body 203 is disposed in the protrusion arrangement portion 66.

[0084] The actuation portion 7 is pressed to the one side D1 in the axis intersecting direction Da by the elastic portion 11. Therefore, the axially facing portion 75 of the actuation portion 7 is located at the initial position Z0. As a result, the protruding portion 204 and the axially facing portion 75 of the plug body 203 face each other in the axial direction AD1 and are in contact with each other.

That is, at the initial position **Z0**, the axially facing portion **75** faces the protruding portion **204** protruding from the outer peripheral surface of the second joint **200** in the axial direction **AD1** from the front end **5a** side of the first joint body **1** in the axial direction **AD1**, in a state where the first joint body **1** is connected to the second joint **200**. Therefore, the coupling state between the first joint **100** and the second joint **200** becomes the locked state.

[0085] On the other hand, when an external force is applied to the actuation portion **71** toward the other side **D2** in the axis intersecting direction **Da**, the axially facing portion **75** moves from the initial position **Z0** to the open position **Z1**. At the open position **Z1**, the axially facing portion **75** does not face the protruding portion **204** in the axial direction **AD1**. As a result, the front side in the axial direction **AD1** of the protruding portion **204** is opened. Therefore, the coupled state between the first joint **100** and the second joint **200** becomes the unlocked state. That is, the coupled state between the first joint **100** and the second joint **200** is unlocked. Therefore, the second joint **200** can be separated from the first joint **100**. The “external force” is applied to the actuation portion **7** (actuation portion **71**) by an operator, for example. For example, the operator presses the actuation portion **71** with a finger.

[0086] As described above with reference to FIG. 7, the actuation portion **71** is actuated along the axis intersecting direction **Da** to switch between the facing state and the non-facing state of the axially facing portion **75** and the protruding portion **204** in the axial direction **AD1**.

[0087] Next, the actuation portion **7** will be described with reference to FIGS. 4, 8, and 9. FIG. 8 is a plan view illustrating the first joint **100**. FIG. 9 is a cross-sectional view taken along line IX-IX of FIG. 8.

[0088] As illustrated in FIGS. 4 and 8, the mounting portion **3** has an intersecting direction facing portion **30**. In the present example embodiment, the intersecting direction facing portion **30** has two opposite pieces **310**. The two opposite pieces **310** face each other in the direction **Db** (FIG. 4).

[0089] As illustrated in FIG. 9, in a state where the axially facing portion **75** and the protruding portion **204** face each other (FIG. 7), the intersecting direction facing portion **30** faces the actuation portion **71** in the axis intersecting direction **Da** between the first joint body **1** and the actuation portion **71**. Therefore, according to the present example embodiment, the movement of the actuation portion **71** in the axis intersecting direction **Da** is restricted by the intersecting direction facing portion **30**. As a result, malfunction of the actuation portion **71** can be reduced or prevented. Therefore, it is possible to prevent the state in which the axially facing portion **75** and the protruding portion **204** face each other from becoming the non-facing state due to the malfunction of the actuation portion **71** and the coupling state between the first joint **100** and the second joint **200** from becoming the unlocked state.

[0090] The actuation portion **71** is pressed to the one side **D1** in the axis intersecting direction **Da** by the elastic portion **11** (FIG. 7). In the state where the axially facing portion **75** and the protruding portion **204** face each other (FIG. 7), the intersecting direction facing portion **30** faces the actuation portion **71** in the axis intersecting direction **Da** between the socket body **50** and the actuation portion **71**. Therefore, the movement of the actuation portion **71** toward the other side **D2** in the axis intersecting direction **Da** is restricted by the intersecting direction facing portion **30**.

[0091] As illustrated in FIG. 9, the mounting portion **3** is located at a first position **P1** in the axial direction **AD1**. The first position **P1** indicates a position where the intersecting direction facing portion **30** faces the actuation portion **71** in the axis intersecting direction **Da**.

[0092] Next, the case where the mounting portion **3** is located at a second position **P2** will be described with reference to FIGS. 10 and 11. FIG. 10 is a plan view illustrating the first joint **100** when the mounting portion **3** is located at the second position **P2**. FIG. 11 is a cross-sectional view taken along line XI-XI of FIG. 10.

[0093] As illustrated in FIGS. 10 and 11, the mounting portion **3** is located at the second position **P2**. The second position **P2** indicates a position where the intersecting direction facing portion **30** is separated from the first position **P1** in the axial direction **AD1**. In other words, the second position

P2 indicates a position where the intersecting direction facing portion **30** is separated rearward in the axial direction AD1 with respect to the first position P1. In other words, the second position P2 indicates a position where the intersecting direction facing portion **30** does not face the actuation portion **71** in the axis intersecting direction Da. Therefore, when the mounting portion **3** is located at the second position P2, the movement of the actuation portion **7** along the axis intersecting direction Da is not restricted. As a result, the operator can release the lock by the axially facing portion **75** and easily separate the second joint **200** from the first joint **100** by pushing the actuation portion **71** to the other side D2 in the axis intersecting direction Da.

[0094] In the present example embodiment, as an example, the position of the mounting portion **3** (the first position P1 and the second position P2) is indicated by the position of the intersecting direction facing portion **30** in the axial direction AD1.

[0095] Next, the movement of the mounting portion **3** will be described with reference to FIGS. **8** to **11**. The mounting portion **3** is movable between the first position P1 and the second position P2 along the axial direction AD1. Therefore, according to the present example embodiment, the operator can easily switch the coupling state between the first joint **100** and the second joint **200** between the locked state and the unlocked state by moving the mounting portion **3** in the axial direction AD1 between the first position P1 and the second position P2.

[0096] In the present example embodiment, as illustrated in FIGS. **9** and **11**, the first joint body **1** includes a first restricting portion **80**. Specifically, the socket body **50** includes the first restricting portion **80**. The first restricting portion **80** restricts the movement of the intersecting direction facing portion **30** between the first position P1 and the second position P2. Therefore, according to the present example embodiment, even when an unexpected force such as vibration is applied to the mounting portion **3**, it is possible to suppress the movement of the mounting portion **3** from the first position P1 to the second position P2. As a result, even when an unexpected force such as vibration is applied to the mounting portion **3**, the movement of the actuation portion **7** in the axis intersecting direction Da is restricted by the intersecting direction facing portion **30**. Therefore, malfunction of the actuation portion **7** is effectively reduced or prevented, and the coupling between the first joint **100** and the second joint **200** can be reduced or prevented from being unintentionally brought into the unlocked state.

[0097] Further, in the present example embodiment, the first restricting portion **80** extends in the axial direction AD1 in a cross-sectional view. The first restricting portion **80** has, for example, a wave shape in a cross-sectional view. In the examples of FIGS. **9** and **11**, the first restricting portion **80** has a first recess **81** and a convex portion **82**. The first recess **81** is recessed inward in the radial direction RD1 with respect to the central axis AX1. The convex portion **82** is connected to the first recess **81**. The convex portion **82** protrudes outward in the radial direction RD1. The convex portion **82** is located farther from the actuation portion **71** in the axial direction AD1 than the first recess **81**. That is, the convex portion **82** is positioned behind the first recess **81** in the axial direction AD1. According to the present example embodiment, the first restricting portion **80** can be realized by a simple configuration such as the first recess **81** and the convex portion **82**. In addition, since the shape of the mounting portion **3** only needs to be matched with the shape of the first restricting portion **80**, the mounting portion **3** having a simple configuration can be realized.

[0098] Specifically, each of the first recess **81** and the convex portion **82** has an annular shape extending in the circumferential direction CD1. As illustrated in FIG. **9**, at the first position P1, a rear end portion **320** in the axial direction AD1 of the mounting portion **3** is fitted into the first recess **81**. The convex portion **82** is adjacent to the first recess **81** in the axial direction AD1. Therefore, it is possible to effectively suppress the movement of the mounting portion **3** from the first position P1 to the second position P2 due to vibration or the like.

[0099] In the present example embodiment, the first restricting portion **80** further includes a second recess **83**. The second recess **83** is recessed inward in the radial direction RD1. The second recess **83** is connected to the convex portion **82**. The second recess **83** is located farther from the actuation

portion **71** in the axial direction **AD1** than the convex portion **82**. That is, the second recess **83** is positioned behind the convex portion **82** in the axial direction **AD1**. The first joint body **1** further includes the overhanging portion **64**. The overhanging portion **64** overhangs outward in the radial direction **RD1** from an apex **VT** of the convex portion **82**. That is, a maximum outer diameter **R1** of the overhanging portion **64** is equal to or larger than a maximum outer diameter **R2** of the convex portion **82** (FIG. 6). The overhanging portion **64** is connected to the second recess **83** in the axial direction **AD1**. The overhanging portion **64** is located farther from the actuation portion **71** in the axial direction **AD1** than the second recess **83**. That is, the overhanging portion **64** is positioned behind the second recess **83** in the axial direction **AD1**. Therefore, according to the present example embodiment, as illustrated in FIG. 11, when the mounting portion **3** is located at the second position **P2**, it is possible to suppress the mounting portion **3** from coming off to the rear side in the axial direction **AD1**.

[0100] Specifically, each of the second recess **83** and the overhanging portion **64** has an annular shape extending in the circumferential direction **CD1**. As illustrated in FIG. 11, at the second position **P2**, the rear end portion **320** in the axial direction **AD1** of the mounting portion **3** is fitted into the second recess **83**. The convex portion **82** is adjacent to the second recess **83** in the axial direction **AD1**. Therefore, it is possible to effectively suppress the movement of the mounting portion **3** from the second position **P2** to the first position **P1** due to vibration or the like.

[0101] Next, another function of the mounting portion **3** will be described with reference to FIGS. 8, 10, 12, and 13. FIG. 12 is a plan view illustrating the mounting portion **3**. FIG. 13 is a perspective view illustrating the mounting portion **3**. As illustrated in FIGS. 12 and 13, the mounting portion **3** further includes a second restricting portion **84**. As illustrated in FIG. 8, in a state where the mounting portion **3** is located at the first position **P1**, the second restricting portion **84** faces both end surfaces **710** in the circumferential direction **CD1** of the actuation portion **71**, in the circumferential direction **CD1**. Therefore, the second restricting portion **84** restricts the movement of the actuation portion **71** in the circumferential direction **CD1**. As a result, according to the present example embodiment, it is possible to suppress the intersecting direction facing portion **30** from not facing the actuation portion **71** in the axis intersecting direction **Da**. As a result, malfunction of the actuation portion **71** can be effectively reduced or prevented.

[0102] Specifically, the second restricting portion **84** has a pair of inclined surfaces **840**. The inclined surface **840** is inclined with respect to the central axis **AX1**. The pair of inclined surfaces **840** faces the central axis **AX1** from the front end **5a** toward the rear end **91a** of the first joint body **1**. Specifically, the pair of inclined surfaces **840** is inclined with respect to the central axis **AX1** so as to approach each other from the front end **5a** toward the rear end **91a** of the first joint body **1**.

[0103] In a state where the mounting portion **3** is located at the first position **P1**, the inclined surface **840** faces the end surface **710** of the actuation portion **71** in the circumferential direction **CD1** and comes into contact with the end surface **710**.

[0104] In particular, in the present example embodiment, as illustrated in FIGS. 8 and 10, in a state where the mounting portion **3** is located at the first position **P1** and a state where the mounting portion **3** is located at the second position **P2**, the second restricting portion **84** faces both end surfaces **710** in the circumferential direction **CD1** of the actuation portion **71** in the circumferential direction **CD1**. Therefore, according to the present example embodiment, even when the mounting portion **3** is located at the second position **P2**, the actuation portion **71** can be prevented from being displaced in the circumferential direction **CD1**. As a result, the operator can easily operate the actuation portion **71**.

[0105] As illustrated in FIG. 10, when the mounting portion **3** is located at the second position **P2**, the inclined surface **840** of the second restricting portion **84** is separated from the end surface **710** of the actuation portion **71** in the circumferential direction **CD1**.

[0106] Next, still another function of the mounting portion **3** will be described with reference to FIGS. 8, 12, and 13. The mounting portion **3** has a pair of inclined surfaces **85**. The inclined surface

85 is inclined with respect to the central axis **AX1**. The pair of inclined surfaces **85** extends with respect to the central axis **AX1** from the front end **5a** toward the rear end **91a** of the first joint body **1**. Therefore, according to the present example embodiment, it is easy for the operator to move the mounting portion **3** from the first position **P1** to the second position **P2**.

[0107] Next, still another function of the mounting portion **3** will be described with reference to FIGS. **8**, **12**, and **13**. The mounting portion **3** has a pair of planes **86**. The pair of planes **86** intersects with the axial direction **AD1**. Therefore, according to the present example embodiment, it is easy for the operator to move the mounting portion **3** from the second position **P2** to the first position **P1**.

[0108] Next, still another function of the mounting portion **3** will be described with reference to FIGS. **8** and **12**. The mounting portion **3** further includes a third recess **88**. The third recess **88** is recessed in the axial direction **AD1**. The elastic portion **11** is disposed to overlap the third recess **88** in the axis intersecting direction **Da** (FIG. **4**). The elastic portion **11** supports the actuation portion **71** in the axis intersecting direction **Da**. Therefore, according to the present example embodiment, it is possible to suppress contact between the elastic portion **11** and the mounting portion **3**. As a result, it is possible to suppress detachment of the elastic portion **11** from the first joint body **1**.

[0109] Specifically, the third recess **88** is recessed from the front end **5a** toward the rear end **91a** of the first joint body **1** in the axial direction **AD1**. The third recess **88** is located between socket body **50** and actuation portion **71**.

[0110] Next, still another function of the mounting portion **3** will be described with reference to FIGS. **4** and **13** to **15**. FIG. **14** is a rear view illustrating the mounting portion **3**. As illustrated in FIGS. **13** and **14**, the mounting portion **3** includes a curved portion **87** and an opening **873**. The curved portion **87** is curved along the circumferential direction **CD1** with respect to the central axis **AX1**. As illustrated in FIG. **4**, the curved portion **87** is fitted to the outer peripheral surface of the first joint body **1** (socket body **50**). Therefore, according to the present example embodiment, the mounting portion **3** can be easily mounted on the first joint body **1**, and the mounting portion **3** can be easily removed from the first joint body **1**. Even when the overhanging portion **64** is provided, the mounting portion **3** can be easily mounted on the first joint body **1** (socket body **50**).

[0111] FIG. **15** is a front view illustrating the first joint **100**. As illustrated in FIG. **15**, in the curved portion **87**, an angle θ formed by one end **871** and the other end **872** of the curved portion **87** in the circumferential direction **CD1** with respect to the central axis **AX1** is 230 degrees or more and 250 degrees or less. Therefore, according to the present example embodiment, the mounting portion **3** can be more easily mounted on the first joint body **1** (socket body **50**), and the mounting portion **3** can be reduced or prevented from being detached from the first joint body **1** (socket body **50**) due to vibration or the like.

[0112] As shown in FIGS. **13** and **14**, the curved portion **87** has a first surface portion **881** and a second surface portion **882**. The first surface portion **881** is located at one end portion of the curved portion **87** in the circumferential direction **CD1**. The first surface portion **881** extends along the axis intersecting direction **Da**. The second surface portion **882** is located at the other end portion of the curved portion **87** in the circumferential direction **CD1**. The second surface portion **882** extends along the axis intersecting direction **Da**. The first surface portion **881** and the second surface portion **882** are parallel to each other. Therefore, according to the present example embodiment, since the opening **873** of the curved portion **87** is not curved and the width does not change, the mounting portion **3** can be easily mounted on the first joint body **1** (socket body **50**).

Modification

[0113] A first joint **100A** according to a modification of the first example embodiment will be described with reference to FIGS. **16** to **19**. The present modification is mainly different from the first example embodiment in that the mounting portion **3A** rotates in the circumferential direction **CD1**. The different point between the present modification and the first example embodiment will mainly be described below.

[0114] FIG. 16 is a perspective view illustrating the first joint **100A** according to a modification of the first example embodiment. As illustrated in FIG. 16, the first joint **100A** includes a mounting portion **3A**. The mounting portion **3A** is mounted on the first joint body **1** (socket body **50**). The mounting portion **3A** is curved along the circumferential direction **CD1**. One end **21** and the other end **22** of the mounting portion **3A** in the circumferential direction **CD1** face each other in the circumferential direction **CD1**. The one end **21** and the other end **22** of the mounting portion **3A** are located at a distance in the circumferential direction **CD1**. Therefore, the mounting portion **3A** has a space **SP** between the one end **21** and the other end **22** of the mounting portion **3A**.

[0115] FIG. 17A is a view illustrating a first position **P1** of the mounting portion **3A**. FIG. 17B is a view illustrating a second position **P2** of the mounting portion **3A**. FIG. 18 is a diagram illustrating the first joint **100A** when the mounting portion **3A** is located at the first position **P1**. FIG. 19 is a diagram illustrating the first joint **100A** when the mounting portion **3A** is located at the second position **P2**.

[0116] As illustrated in FIGS. 17A and 17B, the mounting portion **3A** is rotatable between the first position **P1** and the second position **P2** along the circumferential direction **CD1** with respect to the central axis **AX1**. The first position **P1** indicates a position where the intersecting direction facing portion **30** faces the actuation portion **71** in the axis intersecting direction **Da**. As illustrated in FIG. 18, at the first position **P1**, the actuation portion **71** and the intersecting direction facing portion **30** face each other in the axis intersecting direction **Da**. Therefore, the intersecting direction facing portion **30** restricts the movement of the actuation portion **71** along the axis intersecting direction **Da**. As a result, malfunction of the actuation portion **7** can be reduced or prevented. The second position **P2** indicates a position where the intersecting direction facing portion **30** is separated from the first position **P1** in the circumferential direction **CD1**. As illustrated in FIG. 19, at the second position **P2**, the actuation portion **71** faces the space **SP** in the axis intersecting direction **Da**. Therefore, the movement of the actuation portion **7** along the axis intersecting direction **Da** is not restricted. As a result, the operator can easily release the lock by pushing the actuation portion **7** to the other side **D2** in the axis intersecting direction **Da**. The first position **P1** and the second position **P2** are separated from each other in the circumferential direction **CD1** with respect to the central axis **AX1**.

[0117] As described above with reference to FIGS. 16 to 19, according to the present modification, the coupling state between the first joint **100A** and the second joint **200** can be easily switched between the locked state and the unlocked state by rotating the mounting portion **3A** in the circumferential direction **CD1**.

Second Example Embodiment

[0118] A first joint **2100** according to a second example embodiment of the present disclosure will be described with reference to FIGS. 20 to 22. FIGS. 20 and 21 are cross-sectional views illustrating the first joint **2100** according to the second example embodiment. FIGS. 20 and 21 illustrate a state in which the first joint **2100** is separated from the second joint **200**. FIG. 20 illustrates an actuation portion **2007** when located at an initial position **Z0**. FIG. 21 illustrates the actuation portion **2007** when located at an open position **Z1**.

[0119] In the first joint **100** according to the first example embodiment, the intersecting direction facing portion **30** of the mounting portion **3** faces the actuation portion **71** in the axis intersecting direction **Da** between the first joint body **1** and the actuation portion **71**, whereas in the first joint **2100** according to the second example embodiment, a mounting portion **2003** has a mounting portion protruding portion **2003a** disposed in a hole **2071a** of an actuation portion **2071**.

Hereinafter, in the second example embodiment, matters different from those in the first example embodiment will be described, and the duplicated part of the first example embodiment will not be described.

[0120] The first joint **2100** includes a first joint body **2001**, the mounting portion **2003**, the actuation portion **2071**, and an elastic portion **2011**. Specifically, the first joint **2100** includes an

actuation portion **2007**. The actuation portion **2007** includes the actuation portion **2071**. The actuation portion **2071** has a hole **2071a**. The hole **2071a** only needs to extend along the axial direction **AD1** and may or may not penetrate the actuation portion **2071**, and has, for example, a cylindrical shape.

[0121] The first joint body **2001** extends along the axial direction **AD1**. The first joint body **2001** is coupled to the second joint **200** along the central axis **AX1**. The mounting portion **2003** is mounted on the outer peripheral surface of the first joint body **2001**. The actuation portion **2007** is operated by an operator to switch the coupling state between the first joint body **2001** and the second joint **200** between the locked state and the unlocked state. The elastic portion **2011** supports the actuation portion **2007**. As an example, the actuation portion **2007** (actuation portion **2071**) and the elastic portion **2011** constitute a push button.

[0122] Specifically, the first joint body **2001** includes a socket part **2005** and an insertion part **9**. The socket part **2005** has a socket body **2050**.

[0123] With continued reference to FIGS. **20** and **21**, the configuration related to the actuation portion **2007** and the actuation portion **2007** will be described. The socket body **2050** further includes accommodating portion **2052**. In the accommodating portion **2052**, a through hole is provided on one side **D1** of the socket body **2050** in the axis intersecting direction **Da**, and the inner peripheral surface on the other side **D2** of the socket body **2050** in the axis intersecting direction **Da** is recessed in the axis intersecting direction **Da**. The actuation portion **2007** is disposed in the accommodating portion **2052** along the axis intersecting direction **Da**. The elastic portion **2011** is disposed in the accommodating portion **2052** along the axis intersecting direction **Da**. The elastic portion **2011** has elasticity. The elastic portion **2011** is, for example, a compression coil spring. The elastic portion **2011** extends and contracts in the axis intersecting direction **Da**. The elastic portion **2011** is disposed between the actuation portion **2007** and a bottom surface **2052a** of the accommodating portion **2052**. The elastic portion **2011** supports the actuation portion **2007**.

[0124] The actuation portion **2007** further includes a pair of connecting portions (not illustrated) and an axially facing portion **2075**. The pair of connecting portions connects the actuation portion **2071** and the axially facing portion **2075**.

[0125] In a state where the actuation portion **2007** is disposed in the accommodating portion **2052**, the elastic portion **2011** pushes the actuation portion **2071** toward the one side **D1** in the axis intersecting direction **Da**. At the initial position **Z0**, the axially facing portion **2075** faces the protrusion arrangement portion **66** in the axial direction **AD1**. That is, at the initial position **Z0**, the axially facing portion **2075** is located in front of the protrusion arrangement portion **66** in the axial direction **AD1**. Therefore, the coupling state between the first joint **2100** and the second joint **200** becomes the locked state.

[0126] On the other hand, when a force is applied to the actuation portion **2007** toward the other side **D2** in the axis intersecting direction **Da** in a state where the actuation portion **2007** is disposed in the accommodating portion **2052**, the actuation portion **2007** moves from the initial position **Z0** toward the other side **D2** in the axis intersecting direction **Da** against the elastic force of the elastic portion **2011**. As a result, the axially facing portion **2075** moves from the initial position **Z0** to the open position **Z1** and stands still. At the open position **Z1**, the axially facing portion **2075** is located on the other side **D2** in the axis intersecting direction **Da** with respect to the protrusion arrangement portion **66**. Therefore, at the open position **Z1**, the axially facing portion **2075** does not face the protrusion arrangement portion **66** in the axial direction **AD1**. As a result, the front side in the axial direction **AD1** of the protrusion arrangement portion **66** is opened at the open position **Z1**. Therefore, the coupling state between the first joint **2100** and the second joint **200** becomes the unlocked state.

[0127] When the force on the actuation portion **2007** is released, due to the elastic force of the elastic portion **2011**, the actuation portion **2007** moves toward the one side **D1** in the axis intersecting direction **Da**. That is, the actuation portion **2007** returns to the initial position **Z0**.

[0128] As described above with reference to FIGS. 20 and 21, the actuation portion 2071 is actuated along the axis intersecting direction Da to switch between the facing state and the non-facing state of the axially facing portion 2075 and the protruding portion 204 in the axial direction AD1.

[0129] The first joint 2100 according to the second example embodiment will be described with reference to FIGS. 20 to 22. FIG. 22 is a cross-sectional view illustrating the first joint 2100 according to the second example embodiment. FIG. 22 illustrates a state in which the first joint 2100 is separated from the second joint 200. FIG. 22 illustrates the actuation portion 2007 when located at the initial position Z0.

[0130] As illustrated in FIGS. 20 to 22, the mounting portion 2003 has a mounting portion protruding portion 2003a. The mounting portion protruding portion 2003a is an example of a mounting portion restricting portion. The shape of the mounting portion protruding portion 2003a is not particularly limited as long as it can be inserted into the hole 2071a, and is, for example, a cylindrical body.

[0131] The mounting portion protruding portion 2003a is disposed in the hole 2071a of the actuation portion 2071 in a state where the axially facing portion 2075 and the protruding portion 204 face each other. Therefore, according to the second example embodiment, the movement of the actuation portion 2071 in the axis intersecting direction Da is restricted by the mounting portion protruding portion 2003a. As a result, malfunction of the actuation portion 2071 can be reduced or prevented. Therefore, it is possible to prevent a state in which the axially facing portion 2075 and the protruding portion 204 face each other from becoming the non-facing state due to the malfunction of the actuation portion 2071 and the coupling state between the first joint 2100 and the second joint 200 from becoming the unlocked state.

[0132] The mounting portion 2003 is located at the first position P1 in the axial direction AD1. The first position P1 indicates a position where the mounting portion protruding portion 2003a is disposed in the hole 2071a. The first position P1 indicates the position of the mounting portion 2003 in FIG. 22.

[0133] On the other hand, the mounting portion 2003 is located at the second position P2. The second position P2 indicates a position where the mounting portion protruding portion 2003a is separated from the first position P1 in the axial direction AD1. In other words, the second position P2 indicates a position where the mounting portion protruding portion 2003a is separated rearward in the axial direction AD1 with respect to the first position P1. In other words, the second position P2 indicates a position where the mounting portion protruding portion 2003a is not disposed in the hole 2071a. The second position P2 indicates the position of the mounting portion 2003 in FIGS. 20 and 21. Therefore, when the mounting portion 2003 is located at the second position P2, the movement of the actuation portion 2007 along the axis intersecting direction Da is not restricted. As a result, the operator can release the lock by the axially facing portion 2075 and easily separate the second joint 200 from the first joint 2100 by pushing the actuation portion 2071 to the other side D2 in the axis intersecting direction Da.

[0134] In the second example embodiment, as an example, the position of the mounting portion 2003 (the first position P1 and the second position P2) is indicated by the position of the mounting portion protruding portion 2003a in the axial direction AD1.

[0135] Next, movement of the mounting portion 2003 will be described. The mounting portion 2003 is movable between the first position P1 and the second position P2 along the axial direction AD1. Therefore, according to the second example embodiment, the operator can easily switch the coupling state between the first joint 2100 and the second joint 200 between the locked state and the unlocked state by moving the mounting portion 2003 in the axial direction AD1 between the first position P1 and the second position P2.

Third Example Embodiment

[0136] A first joint 3100 according to a third example embodiment of the present disclosure will be

described with reference to FIGS. 23 and 24. FIGS. 23 and 24 are cross-sectional views illustrating the first joint 3100 according to the third example embodiment. FIGS. 23 and 24 illustrate a state in which the first joint 3100 is separated from the second joint 200. FIG. 23 illustrates the actuation portion 3007 when located at the initial position Z0, and the mounting portion 3003 when located at the first position P1. FIG. 24 illustrates the actuation portion 3007 when located at the open position Z1 and the mounting portion 3003 when located at the second position P2.

[0137] In the first joint 2100 according to the second example embodiment, the mounting portion 2003 has the mounting portion protruding portion 2003a disposed in the hole 2071a of the actuation portion 2071. On the other hand, in the first joint 3100 according to the third example embodiment, the mounting portion 3003 has a cover portion 3003a disposed on the outer side in the radial direction RD1 with respect to the surface on the outer side in the radial direction RD1 of the actuation portion 3071. Hereinafter, in the third example embodiment, matters different from those in the second example embodiment will be described, and the duplicated part of the second example embodiment will not be described.

[0138] The first joint 3100 includes a first joint body 3001, a mounting portion 3003, an actuation portion 3071, and an elastic portion 3011. Specifically, the first joint 3100 includes an actuation portion 3007. The actuation portion 3007 has the actuation portion 3071.

[0139] The first joint body 3001 extends along the axial direction AD1. The first joint body 3001 is coupled to the second joint 200 along the central axis AX1. The mounting portion 3003 is mounted on the outer peripheral surface of the first joint body 3001. The actuation portion 3007 is operated by an operator to switch the coupling state between the first joint body 3001 and the second joint 200 between the locked state and the unlocked state. The elastic portion 3011 supports the actuation portion 3007. As an example, the actuation portion 3007 (actuation portion 3071) and the elastic portion 3011 constitute a push button.

[0140] Specifically, the first joint body 3001 includes a socket part 3005 and an insertion part 9. The socket part 3005 has a socket body 3050.

[0141] With continued reference to FIGS. 23 and 24, the configuration related to the actuation portion 3007 and the actuation portion 3007 will be described. The socket body 3050 further includes an accommodating portion 3052. In the accommodating portion 3052, a through hole is provided on one side D1 of the socket body 3050 in the axis intersecting direction Da, and the inner peripheral surface on the other side D2 of the socket body 3050 in the axis intersecting direction Da is recessed in the axis intersecting direction Da. The actuation portion 3007 is disposed in the accommodating portion 3052 along the axis intersecting direction Da. The elastic portion 3011 is disposed in the accommodating portion 3052 along the axis intersecting direction Da. The elastic portion 3011 is disposed between the actuation portion 3007 and a bottom surface 3052a of the accommodating portion 3052.

[0142] The actuation portion 3007 further includes a pair of connecting portions (not illustrated) and an axially facing portion 3075. The pair of connecting portions connects the actuation portion 3071 and the axially facing portion 3075.

[0143] In a state where the actuation portion 3007 is disposed in the accommodating portion 3052, the elastic portion 3011 pushes the actuation portion 3071 toward the one side D1 in the axis intersecting direction Da. At the initial position Z0, the axially facing portion 3075 faces the protrusion arrangement portion 66 in the axial direction AD1. That is, at the initial position Z0, the axially facing portion 3075 is located in front of the protrusion arrangement portion 66 in the axial direction AD1. Therefore, the coupling state between the first joint 3100 and the second joint 200 becomes the locked state. At the initial position Z0, the outer surface of the actuation portion 3007 in the radial direction RD1 is substantially flush with the outer surface of the first joint body 3001 in the radial direction RD1. Note that the outer surface in the radial direction RD1 of the actuation portion 3007 may be disposed on the inner side in the radial direction RD1 with respect to the outer surface in the radial direction RD1 of the first joint body 3001.

[0144] On the other hand, when a force is applied to the actuation portion **3007** toward the other side **D2** in the axis intersecting direction **Da** in a state where the actuation portion **3007** is disposed in the accommodating portion **3052**, the actuation portion **3007** moves from the initial position **Z0** toward the other side **D2** in the axis intersecting direction **Da** against the elastic force of the elastic portion **3011**. As a result, the axially facing portion **3075** moves from the initial position **Z0** to the open position **Z1** and stands still. At the open position **Z1**, the axially facing portion **3075** is located on the other side **D2** in the axis intersecting direction **Da** with respect to the protrusion arrangement portion **66**. Therefore, at the open position **Z1**, the axially facing portion **3075** does not face the protrusion arrangement portion **66** in the axial direction **AD1**. As a result, the front side in the axial direction **AD1** of the protrusion arrangement portion **66** is opened at the open position **Z1**. Therefore, the coupling state between the first joint **3100** and the second joint **200** becomes the unlocked state.

[0145] When the force on the actuation portion **3007** is released, due to the elastic force of the elastic portion **3011**, the actuation portion **3007** moves toward the one side **D1** in the axis intersecting direction **Da**. That is, the actuation portion **3007** returns to the initial position **Z0**.

[0146] As described above with reference to FIGS. **23** and **24**, the actuation portion **3071** is actuated along the axis intersecting direction **Da** to switch between the facing state and the non-facing state of the axially facing portion **3075** and the protruding portion **204** in the axial direction **AD1**.

[0147] As illustrated in FIGS. **23** and **24**, the mounting portion **3003** includes a cover portion **3003a**. The cover portion **3003a** is an example of a mounting portion restricting portion. The cover portion **3003a** is, for example, a plate-like body.

[0148] The cover portion **3003a** is disposed on the outer side in the radial direction **RD1** with respect to the outer surface in the radial direction **RD1** of the actuation portion **3071** in the state where the axially facing portion **3075** and the protruding portion **204** face each other. That is, the cover portion **3003a** covers the outer surface in the radial direction **RD1** of the actuation portion **3071**. Therefore, according to the third example embodiment, contact with the actuation portion **3071** is inhibited by the cover portion **3003a**. In other words, according to the third example embodiment, the cover portion **3003a** restricts the movement of the actuation portion **3071** in the axis intersecting direction **Da**. As a result, malfunction of the actuation portion **3071** can be reduced or prevented. Therefore, it is possible to prevent a state in which the axially facing portion **3075** and the protruding portion **204** face each other from becoming the non-facing state due to the malfunction of the actuation portion **3071** and the coupling state between the first joint **3100** and the second joint **200** from becoming the unlocked state.

[0149] The mounting portion **3003** is located at the first position **P1** in the axial direction **AD1**. The first position **P1** indicates a position where the cover portion **3003a** faces the actuation portion **3071** in the axis intersecting direction **Da**. The first position **P1** indicates the position of the mounting portion **3003** in FIG. **23**.

[0150] On the other hand, the mounting portion **3003** is located at the second position **P2**. The second position **P2** indicates a position where the cover portion **3003a** is separated from the first position **P1** in the axial direction **AD1**. In other words, the second position **P2** indicates a position where the cover portion **3003a** is separated rearward in the axial direction **AD1** with respect to the first position **P1**. In other words, the second position **P2** indicates a position where the cover portion **3003a** does not face the actuation portion **3071** in the axis intersecting direction **Da**. The second position **P2** indicates the position of the mounting portion **3003** in FIG. **24**. Therefore, when the mounting portion **3003** is located at the second position **P2**, contact with the actuation portion **3007** is not hindered. As a result, the operator can release the lock by the axially facing portion **3075** and easily separate the second joint **200** from the first joint **3100** by pushing the actuation portion **3071** to the other side **D2** in the axis intersecting direction **Da**.

[0151] In the third example embodiment, as an example, the position (the first position **P1** and the

second position P2) of the mounting portion **3003** is indicated by the position of the cover portion **3003a** in the axial direction AD1.

[0152] Next, movement of the mounting portion **3003** will be described. The mounting portion **3003** is movable between the first position P1 and the second position P2 along the axial direction AD1. Therefore, according to the third example embodiment, the operator can easily switch the coupling state between the first joint **3100** and the second joint **200** between the locked state and the unlocked state by moving the mounting portion **3003** in the axial direction AD1 along the outer surface in the radial direction RD1 of the socket body **3050** between the first position P1 and the second position P2.

Fourth Example Embodiment

[0153] A first joint **4100** according to a fourth example embodiment of the present disclosure will be described with reference to FIGS. 25 to 27. FIGS. 25 and 26 are cross-sectional views illustrating the first joint **4100** according to the fourth example embodiment. FIGS. 25 and 26 illustrate a state in which the first joint **4100** is separated from the second joint **200**. FIG. 25 illustrates an actuation portion **4007** when located at the initial position Z0, and a mounting portion **4003** when located at the first position P1. FIG. 26 illustrates the actuation portion **4007** when located at the open position Z1, and the mounting portion **4003** when located at the second position P2.

[0154] In the first joint **3100** according to the third example embodiment, the inner surface in the radial direction RD1 of the cover portion **3003a** and the outer surface in the radial direction RD1 of the actuation portion **3071** are in contact with each other at the first position P1, whereas in the first joint **4100** according to the fourth example embodiment, the inner surface in the radial direction RD1 of the cover portion **4003a** and the outer surface in the radial direction RD1 of an actuation portion **4071** face each other with a space therebetween at the first position P1. Hereinafter, in the fourth example embodiment, matters different from those in the third example embodiment will be described, and the duplicated part of the third example embodiment will not be described.

[0155] The first joint **4100** includes a first joint body **4001**, the mounting portion **4003**, the actuation portion **4071**, and an elastic portion **4011**. Specifically, the first joint **4100** includes the actuation portion **4007**. The actuation portion **4007** has the actuation portion **4071**.

[0156] The first joint body **4001** extends along the axial direction AD1. The first joint body **4001** is coupled to the second joint **200** along the central axis AX1. The mounting portion **4003** is mounted on the outer peripheral surface of the first joint body **4001**. The actuation portion **4007** is operated by an operator to switch the coupling state between the first joint body **4001** and the second joint **200** between the locked state and the unlocked state. The elastic portion **4011** supports the actuation portion **4007**. As an example, the actuation portion **4007** (actuation portion **4071**) and the elastic portion **4011** constitute a push button.

[0157] Specifically, the first joint body **4001** includes a socket part **4005** and an insertion part **9**. The socket part **4005** has a socket body **4050**.

[0158] With continued reference to FIG. 25, the configuration related to the actuation portion **4007** and the actuation portion **4007** will be described. The socket body **4050** further includes an accommodating portion **4052**. In the accommodating portion **4052**, a through hole is provided on one side D1 of the socket body **4050** in the axis intersecting direction Da, and the inner peripheral surface on the other side D2 of the socket body **4050** in the axis intersecting direction Da is recessed in the axis intersecting direction Da. The actuation portion **4007** is disposed in the accommodating portion **4052** along the axis intersecting direction Da. The elastic portion **4011** is disposed in the accommodating portion **4052** along the axis intersecting direction Da. The elastic portion **4011** is disposed between the actuation portion **4007** and a bottom surface **4052a** of the accommodating portion **4052**.

[0159] The actuation portion **4007** further includes a pair of connecting portions (not illustrated) and an axially facing portion **4075**. The pair of connecting portions connects the actuation portion

4071 and the axially facing portion **4075**.

[0160] In a state where the actuation portion **4007** is disposed in the accommodating portion **4052**, the elastic portion **4011** pushes the actuation portion **4071** toward the one side **D1** in the axis intersecting direction **Da**. At the initial position **Z0**, the axially facing portion **4075** faces the protrusion arrangement portion **66** in the axial direction **AD1**. That is, at the initial position **Z0**, the axially facing portion **4075** is located in front of the protrusion arrangement portion **66** in the axial direction **AD1**. Therefore, the coupling state between the first joint **4100** and the second joint **200** becomes a locked state. At the initial position **Z0**, the outer surface in the radial direction **RD1** of the actuation portion **4007** is disposed on the radially outer side of the outer surface in the radial direction **RD1** of the first joint body **4001**.

[0161] On the other hand, when a force is applied to the actuation portion **4007** toward the other side **D2** in the axis intersecting direction **Da** in a state where the actuation portion **4007** is disposed in the accommodating portion **4052**, the actuation portion **4007** moves from the initial position **Z0** toward the other side **D2** in the axis intersecting direction **Da** against the elastic force of the elastic portion **4011**. As a result, the axially facing portion **4075** moves from the initial position **Z0** to the open position **Z1** and stands still. At the open position **Z1**, the axially facing portion **4075** is located on the other side **D2** in the axis intersecting direction **Da** with respect to the protrusion arrangement portion **66**. Therefore, at the open position **Z1**, the axially facing portion **4075** does not face the protrusion arrangement portion **66** in the axial direction **AD1**. As a result, the front side in the axial direction **AD1** of the protrusion arrangement portion **66** is opened at the open position **Z1**. Therefore, the coupling state between the first joint **4100** and the second joint **200** becomes an unlocked state.

[0162] When the force on the actuation portion **4007** is released, due to the elastic force of the elastic portion **4011**, the actuation portion **4007** moves toward the one side **D1** in the axis intersecting direction **Da**. That is, the actuation portion **4007** returns to the initial position **Z0**.

[0163] As described above with reference to FIGS. 25 and 26, the actuation portion **4071** is actuated along the axis intersecting direction **Da** to switch between the facing state and the non-facing state of the axially facing portion **4075** and the protruding portion **204** in the axial direction **AD1**.

[0164] As illustrated in FIGS. 25 and 26, the mounting portion **4003** includes a cover portion **4003a**. The cover portion **4003a** is an example of a mounting portion restricting portion. The cover portion **4003a** is, for example, a plate-like body.

[0165] The cover portion **4003a** is disposed on the outer side in the radial direction **RD1** with respect to the outer surface in the radial direction **RD1** of the actuation portion **4071** in the state where the axially facing portion **4075** and the protruding portion **204** face each other. That is, the cover portion **4003a** covers the outer surface in the radial direction **RD1** of the actuation portion **4071**. Therefore, according to the fourth example embodiment, contact with the actuation portion **4071** is inhibited by the cover portion **4003a**. In other words, according to the fourth example embodiment, the cover portion **4003a** restricts the movement of the actuation portion **4071** in the axis intersecting direction **Da**. As a result, malfunction of the actuation portion **4071** can be reduced or prevented. Therefore, it is possible to prevent a state in which the axially facing portion **4075** and the protruding portion **204** face each other from becoming the non-facing state due to the malfunction of the actuation portion **4071** and the coupling state between the first joint **4100** and the second joint **200** from becoming the unlocked state.

[0166] The mounting portion **4003** is located at the first position **P1** in the axial direction **AD1**. The first position **P1** indicates a position where the cover portion **4003a** faces the actuation portion **4071** in the axis intersecting direction **Da**. Specifically, at the first position **P1**, the inner surface of the cover portion **4003a** in the radial direction **RD1** and the outer surface of the actuation portion **4071** in the radial direction **RD1** face each other with a space therebetween. The first position **P1** indicates the position of the mounting portion **4003** in FIG. 25.

[0167] On the other hand, the mounting portion **4003** is located at the second position P2. The second position P2 indicates a position where the cover portion **4003a** is separated from the first position P1 in the axial direction AD1. In other words, the second position P2 indicates a position where the cover portion **4003a** is separated rearward in the axial direction AD1 with respect to the first position P1. In other words, the second position P2 indicates a position where the cover portion **4003a** does not face the actuation portion **4071** in the axis intersecting direction Da. The second position P2 indicates the position of the mounting portion **4003** in FIG. 26. Therefore, when the mounting portion **4003** is located at the second position P2, contact with the actuation portion **4007** is not hindered. As a result, the operator can release the lock by the axially facing portion **4075** and easily separate the second joint **200** from the first joint **4100** by pushing the actuation portion **4071** to the other side D2 in the axis intersecting direction Da.

[0168] In the fourth example embodiment, as an example, the position (the first position P1 and the second position P2) of the mounting portion **4003** is indicated by the position of the cover portion **4003a** in the axial direction AD1.

[0169] Next, movement of the mounting portion **4003** will be described. The mounting portion **4003** is movable between the first position P1 and the second position P2 along the axial direction AD1. Therefore, according to the fourth example embodiment, the operator can easily switch the coupling state between the first joint **4100** and the second joint **200** between the locked state and the unlocked state by moving the mounting portion **4003** in the axial direction AD1 between the first position P1 and the second position P2. In addition, even when the cover portion **4003a** is bent by an external force, it is possible to suppress contact with the actuation portion **4071** and to suppress malfunction of the actuation portion **4071**.

[0170] Next, another function of the mounting portion **4003** will be described with reference to FIG. 27. FIG. 27 is a cross-sectional view illustrating the mounting portion **4003**. As illustrated in FIG. 27, the mounting portion **4003** further includes a movement restricting portion **4084**. The movement restricting portion **4084** is, for example, a pair of side walls. Each of the pair of side walls extends from an end of the cover portion **4003a** along the axis intersecting direction Da. Specifically, in a state where the mounting portion **4003** is located at the first position P1, the movement restricting portion **4084** faces both end surfaces **4710** in the circumferential direction CD1 of the actuation portion **4071** in the circumferential direction CD1. Therefore, the movement restricting portion **4084** restricts the movement of the mounting portion **4003** in the circumferential direction CD1. As a result, according to the fourth example embodiment, it is possible to suppress the cover portion **4003a** from not facing the actuation portion **4071** in the axis intersecting direction Da. As a result, malfunction of the actuation portion **4071** can be effectively reduced or prevented.

[0171] In addition, in a state where the mounting portion **4003** is located at the first position P1, the movement restricting portion **4084** faces the end surface **4710** of the actuation portion **4071** in the circumferential direction CD1 and comes into contact with the end surface **4710**.

[0172] The example embodiments of the present disclosure are described above with reference to the drawings. However, the present disclosure is not limited to the above example embodiments, and can be implemented in various modes without departing from the gist of the present disclosure. The plurality of elements disclosed in the above example embodiments can be appropriately modified. For example, one element of all elements illustrated in one example embodiment may be added to an element of another example embodiment, or some elements of all elements illustrated in one example embodiment may be eliminated from the example embodiment.

[0173] Further, the drawings schematically illustrate each element mainly in order to facilitate understanding of the disclosure, and the thickness, length, number, interval, and the like of the illustrated elements may be different from the actual ones for convenience of creation of the drawings. The configuration of each element shown in the above example embodiments is an example and is not particularly limited, and it goes without saying that various modifications can be made without substantially departing from the effects of the present disclosure.

[0174] [1] The mounting portion 3 or 3A may be disposed on the second joint 200.

[0175] Note that the present techniques, example embodiments, or modifications thereof, can have any of the configurations described below.

[0176] (1) A first joint having a central axis and extending in an axial direction along the central axis, the first joint including a first joint body to be coupled to a second joint along the central axis, an axially facing portion that opposes a protruding portion protruding from an outer peripheral surface of the second joint in the axial direction from a front end side in the axial direction of the first joint body in a state in which the first joint body is coupled to the second joint, an actuation portion that is connected to the axially facing portion and is actuated along an axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction, and a mounting portion to be mounted on an outer peripheral surface of the first joint body, wherein the mounting portion includes an actuation portion restricting portion to restrict movement of the actuation portion in the facing state of the axially facing portion and the protruding portion.

[0177] (2) A first joint having a central axis and extending in an axial direction along the central axis, the first joint including a first joint body to be coupled to a second joint along the central axis, an axially facing portion that faces a protruding portion protruding from an outer peripheral surface of the second joint in the axial direction from a front end side in the axial direction of the first joint body in a state in which the first joint body is coupled to the second joint, an actuation portion that is connected to the axially facing portion and is actuated along an axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction, and a mounting portion to be mounted on an outer peripheral surface of the first joint body, wherein the mounting portion includes an intersecting direction facing portion that opposes the actuation portion in the axis intersecting direction between the first joint body and the actuation portion in the facing state of the axially facing portion and the protruding portion.

[0178] (3) The first joint according to (2), wherein the mounting portion is movable between a first position and a second position along the axial direction, the first position indicates a position at which the intersecting direction facing portion opposes the actuation portion in the axis intersecting direction, and the second position indicates a position at which the intersecting direction facing portion is separated from the first position in the axial direction.

[0179] (4) The first joint according to (3), wherein the first joint body includes a first restricting portion to restrict movement of the intersecting direction facing portion between the first position and the second position.

[0180] (5) The first joint according to (4), wherein the first restricting portion includes a first recess recessed inward in a radial direction with respect to the central axis, and a convex portion that is connected to the first recess and protrudes outward in the radial direction, and the convex portion is located farther from the actuation portion in the axial direction than the first recess.

[0181] (6) The first joint according to (5), wherein the first restricting portion further includes a second recess recessed inward in the radial direction, the second recess is connected to the convex portion and is located farther from the actuation portion in the axial direction than the convex portion, the first joint body further includes an overhanging portion that is connected to the second recess in the axial direction and is located farther from the actuation portion in the axial direction than the second recess, and a maximum outer diameter of the overhanging portion is equal to or larger than a maximum outer diameter of the convex portion.

[0182] (7) The first joint according to any one of (2) to (6), wherein the mounting portion includes a curved portion curved along a circumferential direction with respect to the central axis, and an opening, and the curved portion is fitted to the outer peripheral surface of the first joint body.

[0183] (8) The first joint according to (7), wherein an angle defined between one end and another end of the curved portion in the circumferential direction with respect to the central axis is about

230 degrees or more and about 250 degrees or less.

[0184] (9) The first joint according to (7) or (8), wherein the curved portion includes a first surface portion located at one end portion of the curved portion in the circumferential direction and extending along the axis intersecting direction, and a second surface portion located at another end portion of the curved portion in the circumferential direction and extending along the axis line intersecting direction, and the first surface portion and the second surface portion are parallel or substantially parallel to each other.

[0185] (10) The first joint according to any one of (3) to (9), wherein the mounting portion further includes a second restricting portion to restrict movement of the actuation portion in the circumferential direction with respect to the central axis, and the second restricting portion opposes both circumferential end surfaces of the actuation portion in the circumferential direction.

[0186] (11) The first joint according to (10), wherein in a state in which the mounting portion is located at the first position and in a state in which the mounting portion is located at the second position, the second restricting portion opposes both of the circumferential end surfaces of the actuation portion in the circumferential direction.

[0187] (12) The first joint according to any one of (2) to (11), wherein the mounting portion includes a pair of inclined surfaces inclined with respect to the central axis, and the pair of inclined surfaces extends with respect to the central axis from the front end toward a rear end of the first joint body.

[0188] (13) The first joint according to any one of (2) to (12), wherein the mounting portion includes a pair of planes intersecting with the axial direction.

[0189] (14) The first joint according to any one of (2) to (13), further including an elastic portion extending along the axis intersecting direction and having elasticity, wherein the mounting portion further includes a third recess recessed in the axial direction, and the elastic portion overlaps the third recess in the axis intersecting direction and supports the actuation portion in the axis intersecting direction.

[0190] (15) The first joint according to (2), wherein the mounting portion is rotatable between a first position and a second position along a circumferential direction with respect to the central axis, the first position indicates a position at which the intersecting direction facing portion faces the actuation portion in the axis intersecting direction, and the second position indicates a position at which the intersecting direction facing portion is separated from the first position in the circumferential direction.

[0191] (16) A pipe joint including the first joint according to any one of (2) to (15), and the second joint.

[0192] (17) The first joint according to (1), wherein the actuation portion has a hole extending along the axial direction, and the mounting portion includes, as the actuation portion restricting portion, a mounting portion protruding portion disposed in the hole in the facing state of the axially facing portion and the protruding portion.

[0193] (18) The first joint according to (1), wherein the mounting portion includes, as the actuation portion restricting portion, a cover portion disposed radially outside a radially outer surface of the actuation portion in the facing state of the axially facing portion and the protruding portion.

[0194] (19) The first joint according to (18), wherein the mounting portion is movable between a first position and a second position along the axial direction, the first position indicates a position at which the cover portion opposes the actuation portion in the axis intersecting direction, the second position indicates a position at which the cover portion is separated from the first position in the axial direction, and at the first position, a radially inner surface of the cover portion and a radially outer surface of the actuation portion face each other with a space.

[0195] (20) The first joint according to (18) or (19), wherein the mounting portion further includes a movement restricting portion to restrict movement of the mounting portion in a circumferential direction with respect to the central axis, the actuation portion is located radially outside a radially

outer surface of the first joint body in the facing state of the axially facing portion and the protruding portion, and the movement restricting portion faces both end surfaces in the circumferential direction of the actuation portion in the circumferential direction in the facing state of the axially facing portion and the protruding portion.

[0196] (21) A pipe joint including the first joint according to (1) or any one of (17) to (20), and the second joint.

[0197] Example embodiments of the present disclosure can be applied to, for example, first joints and pipe joints.

[0198] While example embodiments of the present disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present disclosure. The scope of the present disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A first joint having a central axis and extending in an axial direction along the central axis, the first joint comprising: a first joint body to be coupled to a second joint along the central axis; an axially facing portion that opposes a protruding portion protruding from an outer peripheral surface of the second joint in the axial direction from a front end side in the axial direction of the first joint body in a state in which the first joint body is coupled to the second joint; an actuation portion that is connected to the axially facing portion and is actuated along an axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction; and a mounting portion to be mounted on an outer peripheral surface of the first joint body; wherein the mounting portion includes an actuation portion restricting portion to restrict movement of the actuation portion in the facing state of the axially facing portion and the protruding portion.

2. A first joint having a central axis and extending in an axial direction along the central axis, the first joint comprising: a first joint body to be coupled to a second joint along the central axis; an axially facing portion that opposes a protruding portion protruding from an outer peripheral surface of the second joint in the axial direction from a front end side in the axial direction of the first joint body in a state in which the first joint body is coupled to the second joint; an actuation portion that is connected to the axially facing portion and is actuated along an axis intersecting direction intersecting the central axis to switch between a facing state and a non-facing state of the axially facing portion and the protruding portion in the axial direction; and a mounting portion to be mounted on an outer peripheral surface of the first joint body; wherein the mounting portion includes an intersecting direction facing portion that opposes the actuation portion in the axis intersecting direction between the first joint body and the actuation portion in the facing state of the axially facing portion and the protruding portion.

3. The first joint according to claim 2, wherein the mounting portion is movable between a first position and a second position along the axial direction; the first position indicates a position at which the intersecting direction facing portion opposes the actuation portion in the axis intersecting direction; and the second position indicates a position at which the intersecting direction facing portion is separated from the first position in the axial direction.

4. The first joint according to claim 3, wherein the first joint body includes a first restricting portion to restrict movement of the intersecting direction facing portion between the first position and the second position.

5. The first joint according to claim 4, wherein the first restricting portion includes: a first recess recessed inward in a radial direction with respect to the central axis; and a convex portion that is connected to the first recess and protrudes outward in the radial direction; and the convex portion is located farther from the actuation portion in the axial direction than the first recess.

- 6.** The first joint according to claim 5, wherein the first restricting portion further includes a second recess recessed inward in the radial direction; the second recess is connected to the convex portion and is located farther from the actuation portion in the axial direction than the convex portion; the first joint body further includes an overhanging portion that is connected to the second recess in the axial direction and is located farther from the actuation portion in the axial direction than the second recess; and a maximum outer diameter of the overhanging portion is equal to or larger than a maximum outer diameter of the convex portion.
- 7.** The first joint according to claim 2, wherein the mounting portion includes: a curved portion curved along a circumferential direction with respect to the central axis; and an opening; and the curved portion is fitted to the outer peripheral surface of the first joint body.
- 8.** The first joint according to claim 7, wherein an angle defined between one end and another end of the curved portion in the circumferential direction with respect to the central axis is about 230 degrees or more and about 250 degrees or less.
- 9.** The first joint according to claim 7, wherein the curved portion includes: a first surface portion located at one end portion of the curved portion in the circumferential direction and extending along the axis intersecting direction; and a second surface portion located at another end portion of the curved portion in the circumferential direction and extending along the axis line intersecting direction; and the first surface portion and the second surface portion are parallel or substantially parallel to each other.
- 10.** The first joint according to claim 3, wherein the mounting portion further includes a second restricting portion to restrict movement of the actuation portion in the circumferential direction with respect to the central axis; and the second restricting portion opposes both circumferential end surfaces of the actuation portion in the circumferential direction.
- 11.** The first joint according to claim 10, wherein in a state in which the mounting portion is located at the first position and in a state in which the mounting portion is located at the second position, the second restricting portion opposes both of the circumferential end surfaces of the actuation portion in the circumferential direction.
- 12.** The first joint according to claim 2, wherein the mounting portion includes a pair of inclined surfaces inclined with respect to the central axis; and the pair of inclined surfaces extends with respect to the central axis from the front end toward a rear end of the first joint body.
- 13.** The first joint according to claim 2, wherein the mounting portion includes a pair of planes intersecting with the axial direction.
- 14.** The first joint according to claim 2, further comprising: an elastic portion extending along the axis intersecting direction and having elasticity; wherein the mounting portion further includes a third recess recessed in the axial direction; and the elastic portion overlaps the third recess in the axis intersecting direction and supports the actuation portion in the axis intersecting direction.
- 15.** The first joint according to claim 2, wherein the mounting portion is rotatable between a first position and a second position along a circumferential direction with respect to the central axis; the first position indicates a position at which the intersecting direction facing portion opposes the actuation portion in the axis intersecting direction; and the second position indicates a position at which the intersecting direction facing portion is separated from the first position in the circumferential direction.
- 16.** A pipe joint comprising: the first joint according to claim 2; and the second joint.
- 17.** The first joint according to claim 1, wherein the actuation portion includes a hole extending along the axial direction; and the mounting portion includes, as the actuation portion restricting portion, a mounting portion protruding portion in the hole in the facing state of the axially facing portion and the protruding portion.
- 18.** The first joint according to claim 1, wherein the mounting portion includes, as the actuation portion restricting portion, a cover portion radially outside a radially outer surface of the actuation portion in the facing state of the axially facing portion and the protruding portion.

19. The first joint according to claim 18, wherein the mounting portion is movable between a first position and a second position along the axial direction; the first position indicates a position at which the cover portion opposes the actuation portion in the axis intersecting direction; the second position indicates a position at which the cover portion is separated from the first position in the axial direction; and at the first position, a radially inner surface of the cover portion and a radially outer surface of the actuation portion face each other with a space.

20. The first joint according to claim 18, wherein the mounting portion further includes a movement restricting portion to restrict movement of the mounting portion in a circumferential direction with respect to the central axis; the actuation portion is located radially outside a radially outer surface of the first joint body in the facing state of the axially facing portion and the protruding portion; and the movement restricting portion opposes both end surfaces in the circumferential direction of the actuation portion in the circumferential direction in the facing state of the axially facing portion and the protruding portion.
